# Real-Time Electrochemical Assessment of Soil Health Through Electrochemically Active Biofilm Formation

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Science – Environmental Quality

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## Abstract

Soil degradation, specifically salinization, currently threatens over 10% of global land and cripples the ecosystem's second-largest carbon sink. Conventional soil health indicators rely heavily on physicochemical properties, creating a critical diagnostic blind spot. Standard Electrical Conductivity (EC) meters cannot distinguish between nutrient-dense soil and toxic, saline-stressed soil, often triggering agricultural false positives. Consequently, farmers unknowingly exacerbate land degradation through improper fertilizer application. This research engineered a novel bioelectrochemical system (BES) to resolve this "Salinity Paradox" by decoupling physical conductivity from microbial vitality. The primary objective was to replace superficial EC measurements with real-time quantification of electrochemically active biofilm (EAB) metabolism. A three-electrode BES utilizing a high-surface-area carbon felt working electrode was deployed across healthy, unhealthy, and saline-stressed soil samples. Geobacter sulfurreducens naturally present in the soil acted as the biocatalyst, performing extracellular electron transfer (EET). Electrochemical techniques, including chronoamperometry (CA), cyclic voltammetry (CV), and differential pulse voltammetry (DPV), were utilized to isolate faradaic biological signals from ohmic abiotic noise. The CA trials revealed a critical inverse correlation: while saline soils exhibited high conductivity, their biological current crashed due to metabolic arrest and protein denaturation. Furthermore, CV trials distinguished the faradaic capacitance of healthy biofilms from the highly linear, ohmic resistance of saline soils ($R^2$ = 0.92). DPV conclusively identified the biological origin of the signal via a distinct OmcZ cytochrome redox peak at -0.13V in healthy soil, which disappeared under saline stress. To automate this analysis, an ensemble machine learning and deep learning model (MSHI) was developed, achieving an 87.5% accuracy, 0.909 F1 score, and 1.0 ROC-AUC using just one hour of initial CA data. This BES successfully establishes the first real-time, label-free indicator of microbial recovery capable of penetrating the saline false positive. By shifting from physical to metabolic analysis, this affordable technology enables precision agriculture and targeted remediation efforts. Ultimately, deploying this sensor empowers farmers to intervene before ecosystem collapse, securing global food supplies and preserving vital carbon sinks.

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## Introduction and Goals

Having double the atmosphere carbon storage capacity (~750 Pg C) and triple the biosphere carbon carbon storage capacity (~560 Pg C), soils possess a carbon storage capacity of 1500 Pg of soil organic carbon in 1m depth alone; furthermore, soil is globally the second largest carbon sink, after oceans (~38,000 Pg C) (Levis, 2010; Pütz et al., 2014). Notably, land degradation - driven from deforestation, wildfire, and post mining areas - led to problems like microbial extinction, soil structure collapse, and loss of carbon sequestration capacity in soil; especially, 133 billion tonnes of carbon has been released from soil since humanity began agriculture, while one third of it could have been deposited back into soil if critical land degradation caused by human actions and industrialization didn't take place (Dynarski et al., 2020; Noronha et al., 2025; 2022; Dunne, 2017). Still, agricultural expansion drives nearly 90% of global deforestation, but soil quality deteriorates in deforestated areas, ironically reducing both yields (by up to 30%) and rainfall (FAO, 2021; Dhakane et al., 2024). These statistics directly manifest the significant problems land degradation poses towards the environment and the society. To effectively mitigate these impacts and restore the soil's carbon capture potential, adequate monitoring of soil health is necessary. However, a critical gap exists between the complex and dynamic system of soil's health and the tools currently used to measure them. In the status quo, conventional soil health evaluation depends heavily on physicochemical indicators including pH, organic matter content, and texture (Bünemann et al., 2018). Firstly, these methods are often labor-intensive, destructive, non-real-time, and spatially sparse (Viscarra Rossel et al., 2011). Moreover, while these parameters provide useful superficial data, they fail to capture the dynamic, functional, and real-time aspects of soil biogeochemistry (Lehmann et al., 2020); they especially fail to capture the state of the microbial system, which is crucial in soil's nutrient cycling, carbon sequestration, and structural stability (Fierer, 2017).

Critically, microbial vitality and redox balance - the important engines of soil functionality - are inexplicable in aforementioned conventional soil health assessment methods. There are no standardized methods of evaluating activity of the microbial activity of the soil, and redox-mediated feedback loops that supervise soil's biological self-repair mechanisms. Thus, during early phases of soil degradation, when intervention is still viable, it is extremely difficult to take action to intervene. When physical symptoms such as aridness and erosion is displayed, the microbial infrastructure is often already collapsed, as well as the difficulty level of soil recovery gets exponentially harder and expensive. Furthermore, in an agricultural perspective, this poses a significant risk for the proper growth of their crops as well.

*[Figure 1: Analysis of conventional soil health indicators]*

| Diagnostic Methods/Factors | Real Time | In-situ | Metabolic Specificity | Salinity Resilient |
|---|---|---|---|---|
| Genomic Sequencing (16S rRna) | No | No | No | Yes |
| CO2 Respiration | No | No | Yes | No |
| Enzyme Assays (Dehydrogenase) | No | No | Yes | Yes |
| PLFA Analysis (Biomass) | No | No | Yes | Yes |
| Physicochemical Sensors (EC/pH) | Yes | Yes | No | No |
| Standard NPK Analysis | No | No | No | No |

Notably, a soil can be chemically rich but metabolically unhealthy, or conversely, highly conductive due to salinity yet biologically malfunctioning. As displayed in the comparative analysis of the conventional soil health indicators above, these methods only focus on the static, temporary, quantities for the soil, but are mostly incapable of understanding the full context of the soil, whether that is its metabolic activity or soil salinization. Consequently, through the analysis of traditional soil health indicators, this study identified an unmet need for a real-time, in-situ biosensing platform capable of quantifying the metabolic vitality of the soil.

Specifically, soil salinization, one of the major parts of soil degradation, gives rise to a 'false - positive' when Electrical Conductivity (EC) meters are employed at the field. Multiple literature state its limitation and incapability to distinguish between high salinity and high nutrients for the soil - therefore being a non-specific method to analyze soil health (Corwin and Lesch, 2005). This is because EC meters are influenced by physicochemical conditions of the soil, leading to a cumulative analysis of multiple properties of soil, not focusing on a single property (Qi et al., 2024). To meet this need, this study manipulates the metabolic activity of bacteria within soil, especially Geobacter sulfurreducens, which is a bacteria that is ubiquitous in any soil. These specific bacteria within the soil perform redox reactions, specifically Extracellular Electron Transfer (EET), which couples the intracellular oxidation of organic substrates like acetate to the reduction of an external solid electron acceptor. In this particular experiment, the carbon felt working electrode of the three electrode system will serve the role of such electron acceptor.

On a larger scale, biofilm is a community of microorganisms that adheres to inert or living surface by a self-produced polymeric matrix; they are ubiquitous in natural ecosystems, and they significantly boost the survival of its incorporated microorganisms by rendering them 1000 times more resistant to diverse types of pressure (Wolfaardt et al., 1999). Biofilms are capable of promoting soil nutrient cycling, thus increasing soil fertility and preventing soil degradation (Kayoumu et al., 2025). EAB, a type of biofilm which this research primarily focuses on, is formed on electrodes of MFCs when electroactive taxa transfer electrons with it (Zhuo et al., 2024). Through CA, literature like (Engel et al., 2019) has confirmed that its currents could be employed to understand the trends of biofilm health, EPS maturation, and EET enhancement.

In addition to EAB formation, this study employs EET driven redox reactions. In this particular research, below is the oxidation reaction which takes place at the working electrode, after the injection of the glucose:

$$C_6H_{12}O_6 + 6H_2O \rightarrow 6CO_2 + 24H^+ + 24e^-$$

The electrons generated from the process above travel to the counter electrode. Below is the reduction reaction which takes place at the counter electrode:

$$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$$

Below is the overall redox coupling:

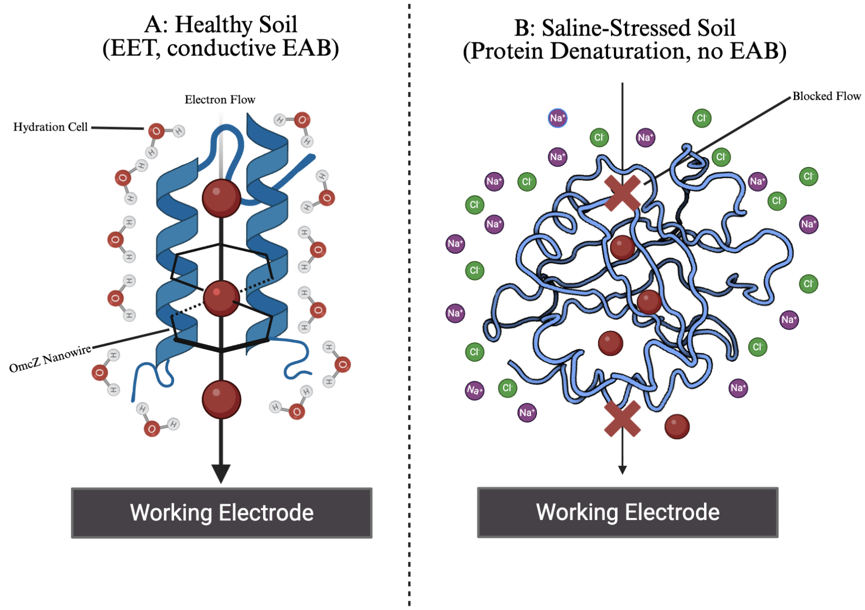
$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$$

After glucose is injected. glucose is converted to fatty acids like acetate by fermentative bacteria, which then serve as a substrate for electrogens like Geobacter, facilitating the EET process. This is because the Geobacter species itself is incapable of directly oxidizing glucose (Freguia et al., 2008). Instead, it consumes the acetate produced by other fermentative bacteria to produce the electrical current measured in this research.

In the phase II of this research, this study specifically applied additional focus on soil salinization as the complexity variable as it currently affects over 1.38 billion hectares of land, which is 10.7% of global land, which causes significant crop losses and threatening food security, while one of its major causes being climate change. Additionally, considering that excessive use of fertilizers, which can come from an inaccurate understanding of a health condition of a soil, being one of the major factors contributing to soil salinization, it is a critical element that is in need for the society.

In the perspective of traditional soil health indicators, soil salinization creates a diagnostic ‘blind spot’, as the high ionic concentration of salt increases the bulk conductivity of the soil, which can potentially demonstrate the similar high linearity and low resistance of a healthy soil. In an electrochemical approach, this distinction can be made because of the electrochemical distinction between the healthy and the saline-stressed soil's mechanism of electron transport. In healthy soil, transport of electrons are faradaic, mediated by the terminal electron conduit of specific Geobacter's outer-membrane cytochromes, like OmcZ (Butler et al., 2010). In saline-stressed soil, high ionic strength gives rise to two cases of failure: the physical denaturation of cytochrome nanowires, and the formation of a dense Helmholtz double layer that electrostatically blocks the electrode. While the soil remains conductive, the saline-stressed soil loses its biological health.

*[Figure 3: Chemical comparison of healthy soil's conductive biofilm with physical denaturation of OmcZ nanowires within saline-stressed soil environment]*




Model A of this figure directly demonstrates the overview of the EET occurring within the soil. EET follows the intracellular metabolism of the Geobacter cell, where one molecule of Acetate is broken down to release 8 electrons, shown in the chemical equation of the oxidation half-reaction:

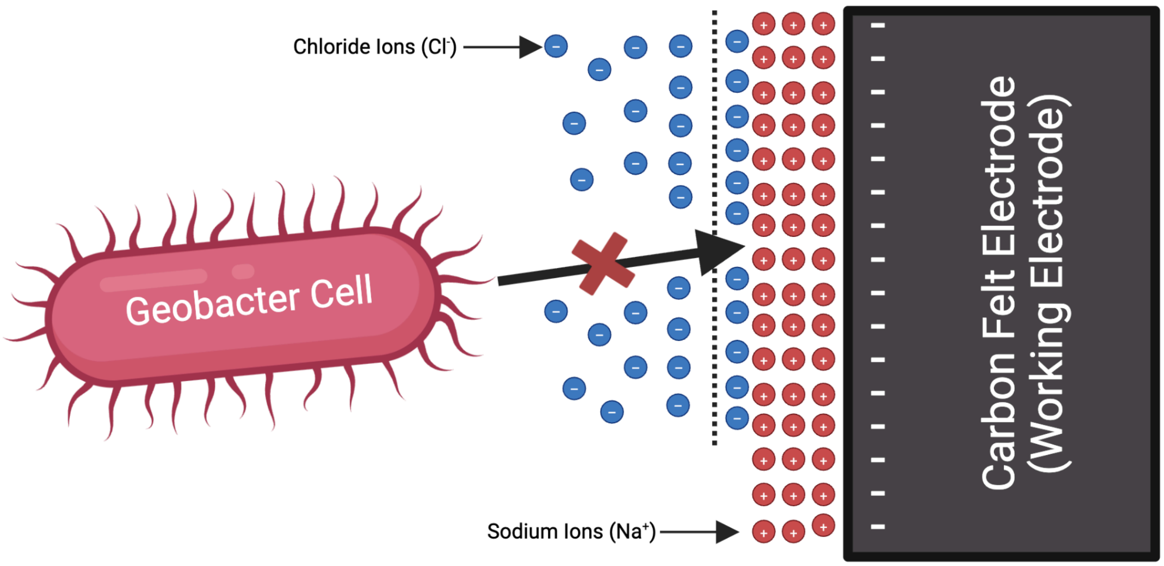
$$CH_3COO^- \ + \ 4H_2O \rightarrow 2HCO_3^- \ + \ 9H^+ \ + \ 8e^-$$

The actual EET reaction happens on OmcZ nanowires. Heme groups (porphyrin rings) are located inside the OmcZ protein, and in the center of every heme group sits a protein-coordinated single Fe atom, which will give rise to distinct redox peaks that can be observed in electrochemical techniques:

$$[Heme - Fe^{3+}] + e^- \rightarrow [Heme - Fe^{2+}]$$

Finally, as these electrons flow through the OmcZ, it approaches the surface of the carbon felt working electrode, which transfers the biological energy to electrical current that the three electrode system monitors.

*[Figure 4: Ionic Shielding at the working electrode surface (Helmholtz Double Layer) in high-salinity environment]*




This study approaches direct, real-time, continuous, non-invasive soil health assessment through the means of electrochemical techniques; additionally, this study extends the line of other papers - which suggests that it indicates soil health via bioelectrochemical signals like CV and CA (Mohamed et al, 2021). This system will measure EET and ionic conduction for the aspect of microbial health and salinization aspect. This study hypothesizes the following: introducing a one-time glucose input and rendering three electrode systems implemented on soil samples (healthy, saline-stressed, and unhealthy soil category) will stimulate the formation of EABs on a carbon felt working electrode, outputting an effective representation of soil health. These electrochemical responses will serve as real-time, label-free indicators of microbial recovery and soil biotic regeneration, demonstrating EAB-based sensing as a novel approach for monitoring soil health.

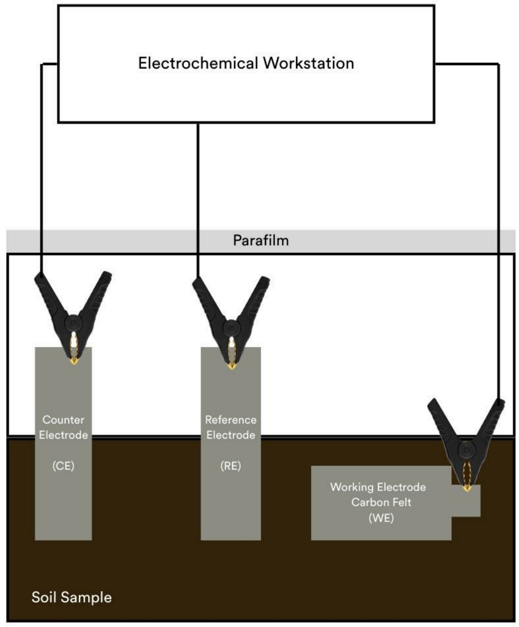
Ultimately, the core environmental crisis addressed in this research is the 'false - positive' paradox created by conventional soil sensors, and to resolve this, this study engineered a bioelectrochemical system to replace superficial physical conductivity readings with a real-time quantification of microbial vitality, connected with a machine learning model trained based on data collected from this research.

## Methodology

There are three categories of soil health incorporated in this research: healthy, unhealthy, and saline-stressed soil category, and all soil samples were collected from the Jilin University Campus Garden. To explicitly define the experimental variables, the autoclaved unhealthy soil served as the negative control group to establish the abiotic baseline. Furthermore, the sodium chloride injection served as the primary independent variable to induce saline-stress and test the sensor's resilience. The collected soil was employed for the healthy soil category of this research, and glucose was added once to all categories to fuel microbial growth, metabolism, and soil respiration (Qi et al., 2022; Zhou et al., 2021). In addition, its injection also had the purpose of experimental acceleration. The stock solution of glucose was prepared by dissolving 18.016g of glucose (0.1 mol) in 100mL of deionized water through a water bath. This yields a concentration of 1.0 molal (m), corresponding to approximately 0.9 Molar (M). This solution and volume (30mL) of glucose will be used always throughout the paper, whenever it mentions the injection of glucose onto the soil sample. It was added only once as adding more would provoke overgrowth, thus it would not align with realistic settings of nutrient pulse that soil typically experiences in the natural ecosystem. Specifically for the saline-stressed soil, sodium chloride was injected to the healthy soil sample and was at rest for 1 hour of dispersion to create a perturbed state of the soil, which will serve as a ‘middle’ health state in between both healthy and unhealthy soil categories. The stock solution of sodium chloride was prepared by dissolving 1.5g of NaCl into 10mL of deionized water through a water bath, yielding a concentration of 0.1M. As the sodium chloride solution is added to the soil sample, the water outside the Geobacter cell is saltier than the water inside, this leads to osmosis, where the water moves outside of the bacterium to balance the pressure caused by the NaCl injection. As a result, the cell creates plasmolysis, shrinking the inner membrane and pulling it away from the outer membrane. Thus, the key c-type cytochromes of the Geobacter cells - MacA (diheme) and PpcA (triheme) - are no longer able to interact, where electron shuttling is included in their interaction. This establishes a state where the bacteria (and its DNA) is maintained, but physically incapable of pushing electrons out. Furthermore, as salt is a stressor, to maintain the water within the cell, bacteria must urgently create osmoprotectants or release the sodium outside of itself. Due to the fact that the bacteria within the soil spends all of their ATP for resisting against sodium rather than focusing on the redox reactions, especially EET.

Notably, the ‘middle’ health state - the saline-stressed state - does not refer to the direct 50% between the healthy and the unhealthy soil category, but rather a state where microorganisms exist within the soil but they are malfunctioning due to the injection of sodium chloride, therefore possibly indicating a false positive state. Another crucial reason that led to this establishment was to spot health conditions of saline soils, which conventional in-situ soil sensors (like TDR or FDR probes), which measures the soil's electrical conductivity, can be mistaken as a fertile soil due to high conductivity and ions (Grisso et al., 2009). While the bulk electrical conductivity correlates with physical parameters like soil texture and water content, it provides no information on microbial metabolic activity or soil respiration. The unhealthy soil category of this experiment was created by putting the collected healthy soil into Boxun's Vertical Pressure Steam Sterilizer at 121°C for 1 hour to thermally kill microorganisms and minerals that the soil sample contained.

*[Figure 5: A diagram of the three-electrode set-up on the soil sample]*




This research consists of two phases of experiments - phase I, where only healthy and unhealthy soil categories were analyzed through electrochemical methods such as CA and CV to initially demonstrate the capabilities of real-time electrochemical assessment of soil health through EAB formation. In addition, pH was measured to ensure that the electrochemical methods did not impact the soil's pH levels, to ensure safety. Phase II employs all three soil categories (healthy, unhealthy, saline-stressed), and they were analyzed through electrochemical methods such as CA, CV, DPV, and OCP, serving as a development of this study based on the findings and analyses of phase I. The objective of phase II was to validate the specificity of the electrochemical assessment method by decoupling ionic conductivity from microbial vitality, therefore confirming the soil's health in a more comprehensive manner. Specifically, this phase aimed to show that the system could distinguish between abiotic noise (high conductivity) and the real biological signal collapse (dysbiosis) in saline-stressed soil samples, in order to consider environmental interference as a factor as well.

In this research, a three-electrode system was employed, and all electrochemical trials were performed on a Chenhua CHI660E electrochemical workstation. For the working electrodes, a 16cm$^2$ (4X4) square electrode was designed using carbon felt; for a counter electrode, CHI129 platinum wire electrode by Shanghai Chenhua was used; and for the reference electrode, a CHI150 saturated calomel electrode by Shanghai Chenhua was used. Carbon felt was used as a working electrode because they are traditionally used in MFC practical applications (Babuta et al., 2012).

## Results

Multiple CA and CV trials were executed for all soil health categories. For phase I, approximately 7 days were spent on both CA and CV trials for all soil health categories, while about 6 days were spent for the unhealthy soil category. For phase II of this research, in addition to CA and CV, electrochemical assessments like OCP and DPV were employed, along with non-electrochemical methods to directly prove the formation of EAB on the working electrode surface.

### Open Circuit Potential Trials

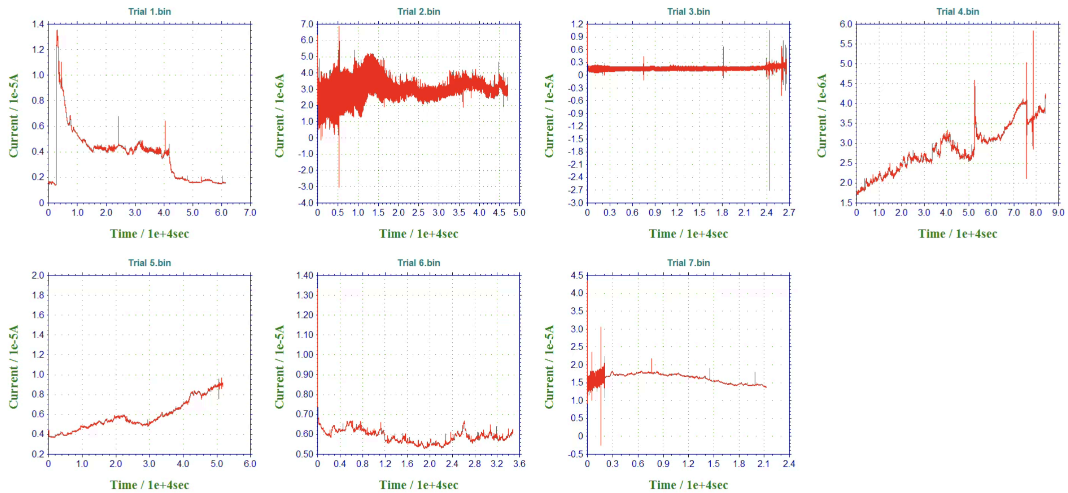
In comparison to the saline-stressed soil category, all trials of the healthy soil category maintained a positive potential. The arithmetic mean of 61mV for the open circuit potentials shows the oxidizing state of the soil environment, which supports aerobic respiration.

The OCP trials of the unhealthy soil show a varied result, as it displays a mix of oxidizing and reducing states. Some of the trials even exhibited a decay/drift, which implies that the ecosystem struggles to maintain the oxidizing environment required for plant health. Starting from OCP trial 2, the data indicates that the soil environment is a reducing environment, displaying a strongly negative potential in comparison to healthy's strongly positive potential. This trial demonstrates the saline-stressed soil undergoing a reductive state, where the high ionic strength from salt injection is preventing the soil from establishing an oxidizing environment which is required for maintaining soil health. For the OCP trials, while the healthy soil displayed stable positive potentials, unhealthy soil displayed mixed or decaying behavior, while saline-stressed soil shifted negative after trial 2, indicating a reducing environment.

### Chronoamperometry Trials

One of the most prominent indicators of EAB formation is secondary rise, as it is the most direct evidence of the EAB growth, surpassing the level of proving its presence. In this study, secondary rise is defined as a rise that is in the shape of a Sigmoid growth curve, involving initial current above 1.0μA, increasing by a significant growth factor of above 15% of the baseline current, and occurs after an initial cottrell decay and a flat line graph. The rationale for such definition is the following: a strict limit of quantification of the initial current was set to be above 1.0μA as useful biocurrents are within the mA and high μA range (Logan et al., 2006). According to Bond & Lovley, electron flux in Geobacter systems is directly proportional to electrode-attached biomass ($R^2 > 0.95$). Therefore, a sustained current increase of 15% threshold serves as a reliable indicator for net metabolic growth, distinguishing active biofilm colonization from passive electrochemical fluctuations (Bond and Lovley, 2003).

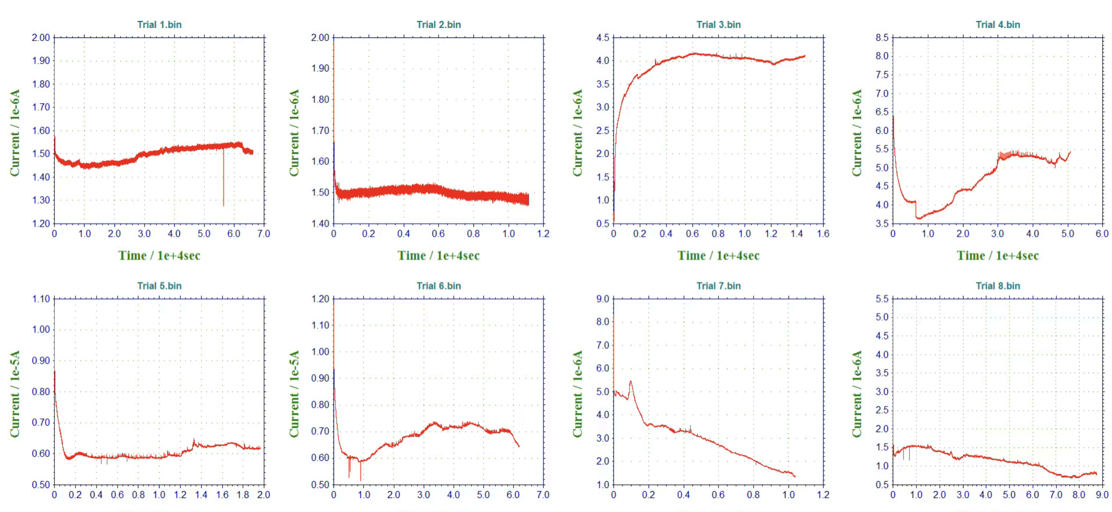
*[Figure 15: Parallel overview of CA trials of healthy soil category (Phase I)]*




Overall, the CA trials of the healthy soil category displayed a steady increase in anodic current density, rising from an initial baseline of approximately 0.125µA/cm2 to a stabilized density at around 0.9375µA/cm2 by the final trial, indicating progressive EAB formation on the surface of carbon felt working electrode. Trial 4 displayed a prominent secondary rise of current (passes all criteria). Trial 5 indicates an even stronger secondary rise, with its current approximately twice as high as trial 4, suggesting increased metabolic activity and improved EET efficiency. Trials 6 and 7 demonstrated biofilm maturation and stabilization.

The most feasible conclusion to make for the unhealthy soil category was that EAB was not formed due to the lack of electroactive taxa that is available in healthy soils. During the unhealthy soil category CA trials, starting from trial 3, the final current started to become negative - which means that the electrons are flowing into the working electrode instead of being dispersed. The healthy soil category for the phase II of this experiment demonstrates a slight secondary rise for trial 4 as the baseline current value is 2.98µA, while the final current is 3.763µA, which is an approximately 126% increase in current.

*[Figure 20: Parallel overview of CA trials of saline-stressed soil category (Phase II)]*



Trial 3 can be interpreted as a chloride corrosion as the current is extremely weak. This shows chloride corrosion because in saline soil environments, salt slowly consumes the electrode surface, then releases a miniscule, constant stream of electrons. Trial 6 actively shows the cell death and the dying saline-stressed soil microbiome. The negative slope of $-1.1*10^{-4}$ indicates either a depletion of electroactive taxa or the passivation of the working electrode from the Helmholtz double layer formation. Overall, this trial confirms the incapabilities of the saline-stressed soil to maintain EET.

**Cyclic Voltammetry Trials**

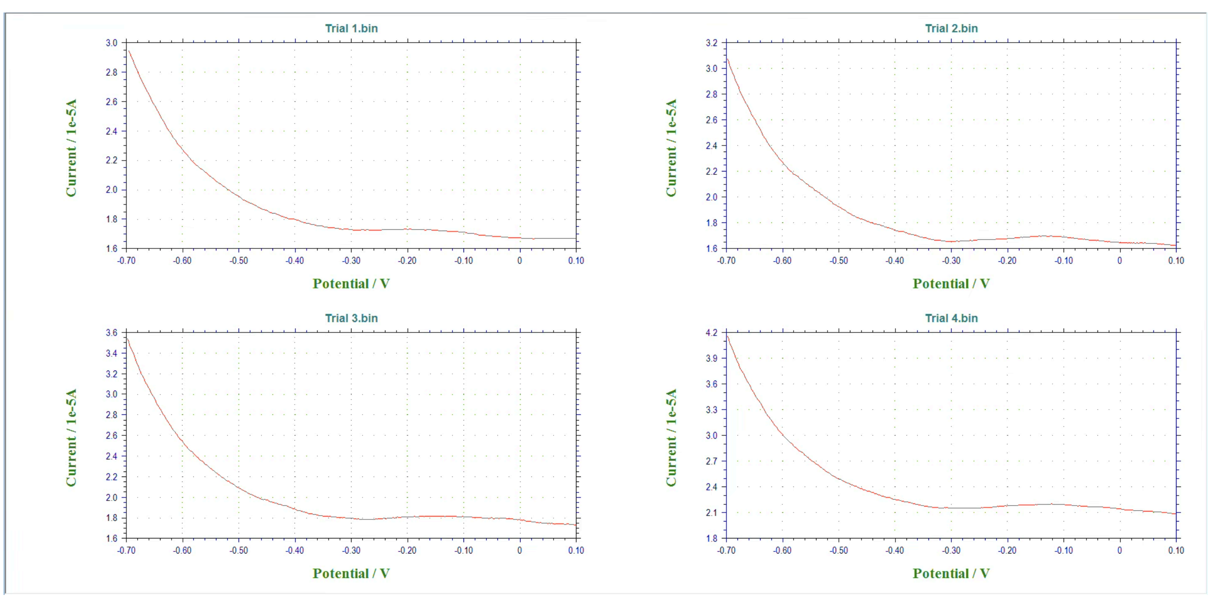
The CV trials of the unhealthy soil category display a highly linear loop, displaying its Ohmic reaction. In comparison to the healthy soil, the initial and final current values are nearly one fourth of it. The saline-stressed soil trials showed capacitive shapes (trial 1), strong linearity/ohmic traces (trials 2-3, $R^2$ = 0.92), higher resistive tilt from Helmholtz double layer (trials 4-5), and cathodic bias from corrosion overpowering anodic biological signal (trial 6).

**$R^2$ of CV Trials**

In electrochemical systems, high $R^2$ implies death, as it confirms that the BES is functioning as an Ohmic resistor, therefore demonstrating the absence of biological activity. In contrast to high $R^2$, low $R^2$ implies life, as living systems are nonlinear and store charge to be released later on, which implies a faradaic system. The unhealthy soil category has an exceptionally higher $R^2$, where each of the $R^2$ value of the unhealthy soil category is larger than the healthy soil category. The value of $R^2$ for the saline-stressed soil category is relatively dynamic, where it even indicates extremely high linearity like 0.924 for CV trial 2 and 3, while there is low $R^2$ like 0.597 and 0.645 for trial 0 and trial 1, respectively. These values well-represent the ambivalence that saline-stress indicates for electroconductive meters.

**Differential Pulse Voltammetry Trials**

*[Figure 75: Parallel overview of DPV trials of healthy soil category]*




The data of the DPV trials show a consistent redox peak centered around -0.13V in the majority of the trials. The peak occurs at -0.136V, -0.144V, -0.120V, for trial 2, 3, and 4 accordingly. This consistency in the redox peak serves as biological evidence of an active cytochrome (most likely OmcZ cytochromes in Geobacter sulfurreducens). This peak strongly suggests the active EET.

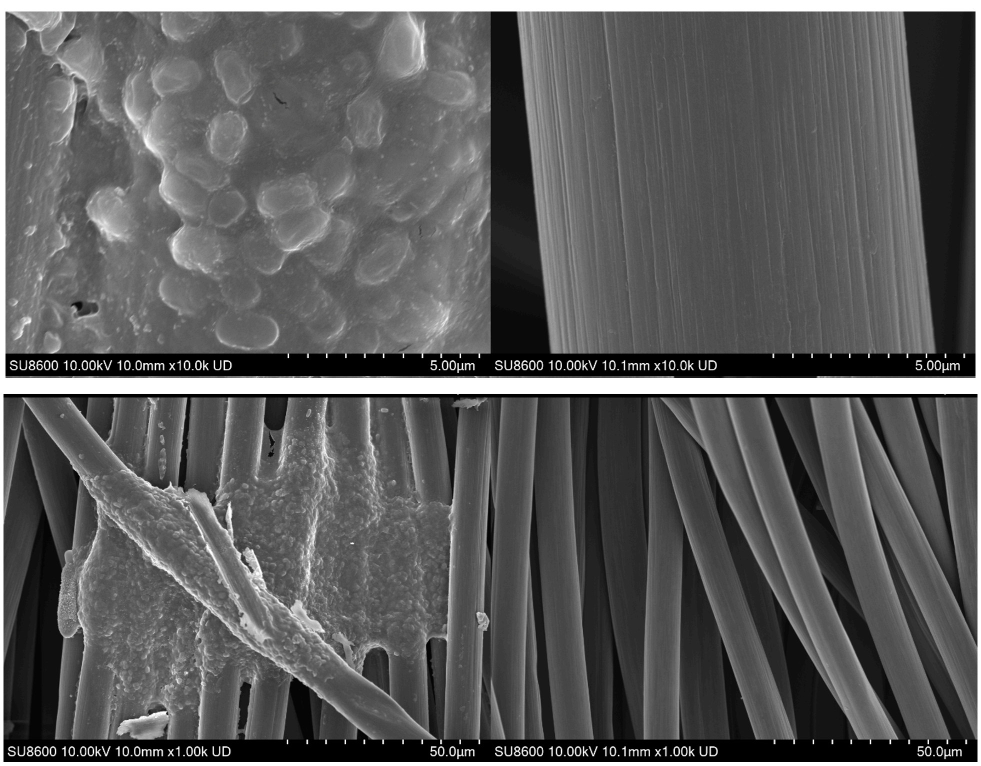
The DPV trials of the unhealthy soil category are prominently unstable. The maximum current was extremely low and consistent (95% drop in signal intensity compared to healthy soil category). The absence of peaks indicates the dormancy of cytochromes and specific proteins that facilitate EET and EAB formation. The saline-stressed soil category DPV trials displayed the loss of this feature. All of these DPV trials show an electricity capacitance, where the current keeps decreasing. There were also small peaks that occurred around 0V, which further reinforces the chloride corrosion cross-verified by other multiple electrochemical assessments.

**pH**

In the healthy soil category, the pH is stable around pH 7.0 (only fluctuating around ±0.1 pH in a neutral pH state for approximately 140 hours). Considering that the US Department of Agriculture National Resources Conservation Services specify an optimal pH range for soil to be [6, 7.5], the pH levels indicated from this category align within this range (USDA NRCS, 2022). The unhealthy soil initial pH was at 7.9, which is out of the aforementioned optimal soil health range. When pH of the soil was tested after 3.6 hours since the initial measurement, it showed 7.3, and was mostly stabilized around 7.0. This implies that the treatment implemented in this study effectively brought the pH down to the optimal pH range. The saline-stressed soil shows minimal changes within the pH of the soil. These results demonstrate the safety of the electrochemical approach and that it is feasible for real world applications.

**Scanning Electron Microscope Images**

***[Figure 81: Parallel comparison of EAB-formed carbon felt and blank carbon felt on a 5μm and 50μm scale (left - EAB-formed, right - blank)]***

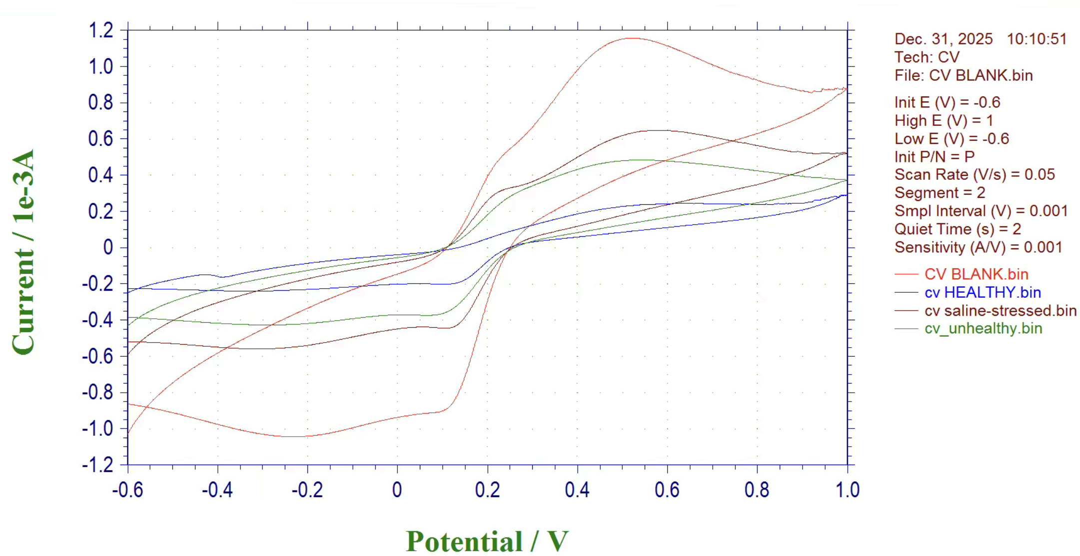


Clearly, the SEM images indicate a significant difference between the working electrode from the healthy soil category, in comparison to a normal carbon felt without any treatment. The images from the 5µm and 10µm scale most effectively displays the EAB formation, showing the colonization of the electroactive taxa on the working electrode surface, covering up the carbon felt surface. The images from the 50µm scale show the EPS largely connecting each microorganism, effectively forming EAB on the carbon felt surface.

**Direct Verification of EAB Formation Via CV and EIS**

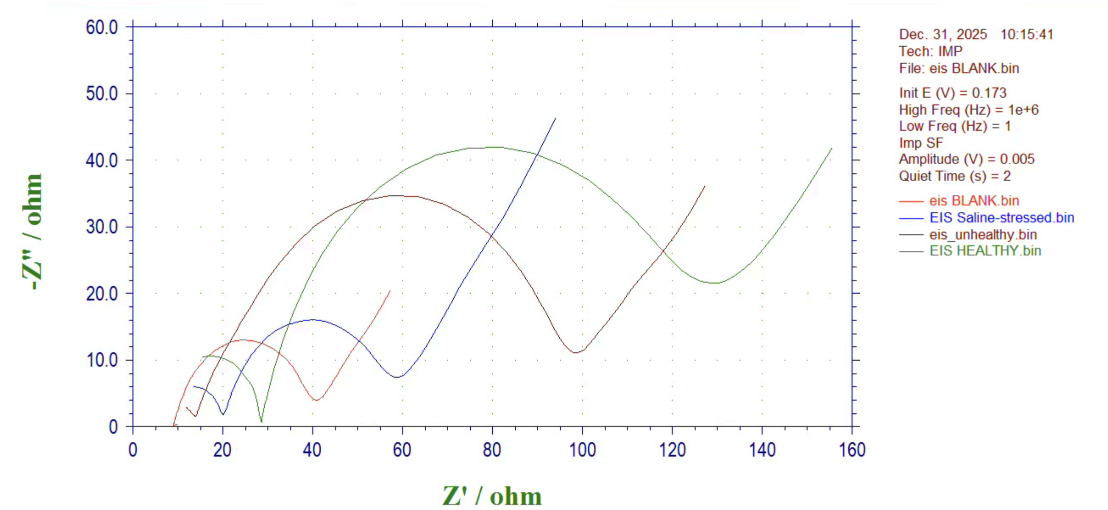
A final CV trial, not on the soil samples, but on the solution containing 20mL of 5 mM $[Fe(CN)_6]^{3-/4-}$ and 0.1 M KCl to verify the direct current and redox peaks of the CV trials of working electrodes from each soil sample; such solution was used based on the verification of working electrode efficiency from other studies (Wenninger et al., 2023). EIS was employed for this study to quantify the resistance of each working electrode from the soil samples in phase II. According to Ohm's law, voltage is proportional to resistance, while current is inversely proportional to resistance. These two electrochemical checks provide one of the strongest validation of the accuracy of this research. Overall, after EAB is formed on the surface of the carbon felt working electrode, the current in the CV trial of the working electrode decreases, and the impedance measured by EIS increases. Due to the fact that microorganisms are non-conductive, they can hinder EET at the working electrode surface.

*[Figure 83: Comparative overlay of CV between carbon felt working electrode from healthy, saline-stressed, unhealthy, and blank category]*




The order of loop size and the redox peaks per category are the following: Blank > Saline-stressed > Unhealthy > Healthy. This is because as biofilm is formed on the working electrode surface, the nonconductive nature of it hinders the electrons to go through the layer, leading to a decreased current per voltage.

*[Figure 84: Comparative overlay of EIS between carbon felt working electrode from healthy, saline-stressed, unhealthy, and blank category]*




The diameter of each semicircle in EIS trials represents the resistance of the carbon felt working electrode. The order of resistance value per each category is the following: Healthy > Unhealthy > Saline-stressed > Blank. The scientific logic is cohesively represented through the CV and EIS trial, considering Ohm's law. As resistance is inversely proportional to current, the reverse order of the results acquired from CV and EIS fully completes the scientific logic established within this study, and effectively confirms the logic of EAB formation across different soil categories.

**Machine Learning & Deep Learning Model Ensemble**

Utilizing the data gained from electrochemical assessments, specifically CA, CV, OCP, DPV from healthy and unhealthy soil category, was used for the training set for establishing a prediction model for a secondary rise using the data of first hour of the CA trial, as well as a Microbial Soil Health Index (MSHI) which indicates the probability of the soil being healthy. MSHI is a soil health index that is established within this study. Saline-stressed was not used for training the model as it ambivalently displays the traits of healthy and unhealthy soil, therefore inadequate for using it to train the model. However, it was used effectively for testing the model and its ability of inference.

MSHI is the main part, where the ensemble occurs between XGBoost (ML) and 1D-CNN (DL) and their index scores are averaged out arithmetically. For XGBoost, 53 metafeatures were defined across CA early features, CA charge and long-term features, CV features, OCP features, and DPV features. The ensemble MSHI model achieved overall accuracy of 87.5% with F1 = 0.909 and ROC-AUC = 1.0. Test samples showed appropriate MSHI values: healthy soils ranged from 0.633 to 0.903, while unhealthy soils ranged from 0.365 to 0.572. Feature importance analysis revealed ca_early_mean (29.8%), ca_early_std (22.3%), ca_early_max (17.4%), ca_early_area_I_dt (11.6%), and cv_loop_area (11.2%) as the top predictive features, demonstrating that the model did not memorize. Saline-stressed soil MSHI inference values clustered around 0.884–0.893. These extensions through model ensemble of DL and ML, and ultimately providing a comprehensive metric that indicates the health of the soil directly helps the farmers as it doesn't require them to analyze the data from the electrochemical trials one by one, where this model can do it for them. This makes the process of soil health analysis more convenient and efficient from the perspective of farmers.

## Discussion

The fundamental objective of this study was a) phase I - differentiate healthy and unhealthy soil in a binary manner, and b) phase II - breakthrough the ‘false-positive’ of saline-stressed soil, and understand a soil's health conditions through employing a wider range of electrochemical assessments to compensate for the superficial analysis made on phase I to differentiate between healthy and unhealthy soil. The BES employed in this study was overall successful by isolating faradaic signals, which is metabolic, from ohmic noise, which is ionic. The OCP trials indicated a distinct thermodynamic difference between healthy and saline-stressed soil, as healthy soil maintained its stable oxidizing potential, which indicates proper metabolism. In direct contrast, the saline-stressed soil displayed a reducing state. This research discovered secondary rise as a major indicator of soil health displayed through CA trials. The healthy soil from phase I (trial 4 & 5) displayed an extremely strong secondary rise, while there was also a weaker presence of secondary rise for healthy soil from phase II (trial 4). As these two soils indicate a similar time frame that displayed secondary rise, this proves the reproducibility of this research in other contexts.

CV trials clearly showed the difference between a capacitive loop and a faradaic loop for healthy and unhealthy soil, and a quantitative analysis was made through calculation of its $R^2$. Saline-stressed soil indicated an extremely high linearity ($R^2 \approx 0.99$) for a few trials, which can be interpreted as its nature of ohmic resistor. This indicates the physical failure of the bacteria, as its protein is denatured. DPV trials indicate the repeated redox peak of -0.12V for the healthy soil, which was attributed to the oxidation of acetate by outer-membrane cytochromes, most likely OmcZ of Geobacter sulfurreducens. In saline-stressed soil trials, a repeated peak occurred at 0V; this demonstrates that soil is detecting the soil's natural chemical noise as the potential of organic matter (humus) is centered around 0V. This further reinforces the notion that the -0.12V indicated from the healthy soil was biological.

In direct comparison to previous literature and commonly held agricultural practices, this study demonstrates that its BES resolves the paradox of the health signal between high salinity and high nutrients. While previous literature, such as the work by Corwin and Lesch, states the profound limitation of apparent soil EC in distinguishing between salinity and fertility, this system overcomes such structural flaws by measuring electron production rate rather than the electron flow path. As salinity suppressed the biological generation of electrons, the sensor successfully displayed a signal drop. This scientific innovation of this research brings significant real world impact in multiple aspects. To begin with, this study scientifically enables precision agriculture for farmers while having the merit of real-time, in-situ, non-destructive, EC sensors, while being capable of analysing soil's metabolic health effectively. The system is self-sustaining through the biochemical nutrient cycle of soil. Additionally, this provides significant benefits to the environment, as proper analysis of soil health prevents farmers to over-fertilize their soil, which was the result of misunderstanding soil health by using the previous methods. This further prevents soil salinization, reduces fertilizer runoff, and reduces eutrophication in waterways, overall providing farmers a higher environmental benefit for their agriculture. On a larger scale, the fact that this study enables a more precise soil remediation effort, this can positively influence climate change mitigation, due to the enhancement of soil's carbon sequestration capabilities. Furthermore, soil salinization costs the world $27.3billion annually. Farmers can save this cost from taking action earlier, rather than waiting for the soil health analysis to arrive or soil to indicate its unhealthiness physically, which renders it expensive to fix the soil. Finally, the model ensemble of machine learning and deep learning provided in this research facilitates the farmers to understand the health of their soil in comprehensive metrics rather than individually analyzing numerous datasets, leading to convenient use of this scientific breakthrough.

## Conclusion and Real-World Applications

This research successfully demonstrated that a three-electrode bioelectrochemical system can provide real-time, non-destructive assessment of soil health through the formation and monitoring of electrochemically active biofilms. The BES was successful in isolating faradaic signals (metabolic) from ohmic noise (ionic), and the system could distinguish between abiotic noise (high conductivity) and the real biological signal collapse (dysbiosis) in saline-stressed soil. The electrochemical responses served as real-time, label-free indicators of microbial recovery and soil biotic regeneration. SEM images and comparative CV/EIS validation directly confirmed EAB formation across soil categories. The pH stability throughout the experiments confirms the safety and feasibility for real-world application. The ensemble MSHI model (XGBoost + 1D-CNN) achieved 87.5% accuracy with ROC-AUC of 1.0, providing a practical automated tool. This scientific innovation provides immediate applications for global agriculture and environmental preservation. Proper analysis of soil health prevents farmers from over-fertilizing their soil, which subsequently reduces fertilizer runoff and minimizes eutrophication in waterways, preventing a chain of environmental side effects that could've occurred otherwise. Furthermore, since soil salinization currently costs the world $27.3 billion annually, farmers can deploy this affordable sensor to take early, preventative action, saving significant remediation costs while preserving the soil's critical carbon sequestration capabilities.

## Limitations and Future Works

Still, there are remaining limitations and challenges that should be considered for future studies and reference. To begin with, the accuracy of the prediction of the ML model might be unstable and weak if assuming applications in wild soil settings, as there may be multiple other noise and field contexts that might be involved in the electrochemical trials that can contribute towards a significant change towards the results. This is because the data gathered through this experiment were lab based, thus it cannot fully consider every aspect of the soil environment. One key factor that was not considered in this research was weather/climate, as it was enclosed in a parafilm limiting influence from temperature and humidity from the external environment. Also, it was not tested in a rainy environment (or other environment apart from the lab environment), which could influence soil texture, structure, noise influence, and pH. It is undeniable that the ecosystem is non-linear and context-dependent; unexpected changes may give rise to significant implications, which were not fully explored in this research. Additionally, it is important to note that there were a total of 5 experiment trials (2 healthy and unhealthy, 1 saline-stressed soil) that was on a single site.

Salinity is also more complex in farms, where agricultural runoffs consist of not only NaCl that was, but also a complex matrix consisting of nitrates, phosphates, and pesticides. Environmental scientists should aim to achieve the following: designing a more powerful working electrode that can overcome the current chloride corrosion. Additionally, it is highly encouraged to focus on DPV trials due to its benefits over the CV trials. On a biological aspect, genetic engineering of bacteria that could replace Geobacter cells to overcome the highly saline environments could be considered. Engineers can focus on the development of screen printed electrodes for affordable, scalable deployment (Mohan et al., 2022). In regards to the modelling ensemble, acquiring more data through more experiments with different soil samples from different areas will fundamentally prevent overfitting and provide diversity to the dataset.

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